

Table 4. Watermark schemes, within-family representative scores, and fixed operating points used in our experiments. Family-level comparisons report multiple methods per family, whereas the Hybrid uses a single pre-specified representative per family. Hyperparameters are held fixed across attacks, with no per-attack retuning. Within each family block, methods are ordered by the absolute score $S(m)$ from highest to lowest.

Family	Method	Experimental Role	Mean robustness (AUROC)	Mean detectability (z -score)	Absolute score $S(m)$	Watermarking Parameters
Biased	Unigram (Zhao et al., 2024)	Fixed-green-list biased sampler	0.812	8.03	0.923	$\gamma = 0.5, \delta = 2.0$
	KGW (Kirchenbauer et al., 2023)	Canonical context-dependent biased sampler	0.800	8.60	0.904	$\gamma = 0.5, \delta = 2.0$, simple-1 seeding
Bias-free	HCW (Hu et al., 2024)	Canonical reweighting design	0.917	3.12	1.160	δ -reweight variant (method=delta)
	DiPMark (Wu et al., 2024)	Token-level bias-free reweighting comparator	0.905	3.60	1.122	$\alpha = 0.45, \gamma = 0.5$
	HeavyWater (Tsur et al., 2025)	Low-entropy bias-free design	0.885	3.78	1.094	Log-normal scores, $k = 1024$, top- $p = 0.999$
	SimplexWater (Tsur et al., 2025)	Low-entropy bias-free design	0.875	4.00	1.075	Simplex-code scores, $ S = m - 1$, top- $p = 0.999$
Semantic	PMark (Huo et al., 2025)	Rejection-based semantic design	0.945	2.12	1.266	Rejection parameter $\rho = 0.25$
	SemStamp (Hou et al., 2024)	LSH-based sentence-level semantic design	0.935	2.43	1.227	SBERT, margin 0.8, $d = 3$, $\gamma = 0.25$
	SimMark (Dabiriaghdam & Wang, 2025)	Similarity-based sentence-level design	0.925	2.62	1.201	Instructor-Large, cosine range [0.68, 0.76], $K = 250$
Distribution-preserving	CGW (Christ et al., 2024)	Canonical distribution-preserving sampler	0.453	-6.63	1.453 only selection	Security parameter $\lambda = 128$, fixed secret key across runs
Training-time baseline	GaussMark (Block et al., 2025)	Structural watermark in model weights	0.835	7.80	0.949 only selection	Gaussian perturbation scale $\sigma = 0.05$ (fixed calibration at $\alpha = 0.05$)

Note. The blue-highlighted rows indicate the representatives used by the Hybrid in the inference-time families. Within each inference-time family, greener score cells indicate higher within-family scores. Under this rule, Unigram is chosen from the biased family, HCW from the bias-free family, and PMark from the semantic family. CGW is the only distribution-preserving candidate. GaussMark is shown separately as the training-time special case outside the Hybrid selector. In the last column, γ denotes the green-list fraction and δ the logit-bias strength in KGW (Kirchenbauer et al., 2023) and Unigram (Zhao et al., 2024). The parameter α in DiPMark (Wu et al., 2024) controls the strength of the bias-free reweighting. In HeavyWater (Tsur et al., 2025), k is the score dimension used to construct the watermark scores. In SimplexWater (Tsur et al., 2025), $|S|$ is the size of the simplex codebook. In SemStamp (Hou et al., 2024), d is the hash dimension. In PMark (Huo et al., 2025), ρ is the rejection parameter. In CGW (Christ et al., 2024), λ is the security parameter. In GaussMark (Block et al., 2025), σ is the standard deviation of the Gaussian perturbation applied to model weights.